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LLM Internal States Reveal Hallucination Risk Faced With a Query

Ziwei Ji, Delong Chen, Etsuko Ishii, Samuel Cahyawijaya,
Yejin Bang, Bryan Wilie, Pascale Fung
Center for Artificial Intelligence Research (CAiRE)
Hong Kong University of Science and Technology
zjiad@connect.ust.hk, pascale@ece.ust.hk

Abstract

The hallucination problem of Large Language Models (LLMs) significantly limits their reliability and trustworthiness. Humans have a self-awareness process that allows us to recognize what we don't know when faced with queries. Inspired by this, our paper investigates whether LLMs can estimate their own hallucination risk before response generation. We analyze the internal mechanisms of LLMs broadly both in terms of training data sources and across 15 diverse Natural Language Generation (NLG) tasks, spanning over 700 datasets. Our empirical analysis reveals two key insights: (1) LLM internal states indicate whether they have seen the query in training data or not; and (2) LLM internal states show they are likely to hallucinate or not regarding the query. Our study explores particular neurons, activation layers, and tokens that play a crucial role in the LLM perception of uncertainty and hallucination risk. By a probing estimator, we leverage LLM self-assessment, achieving an average hallucination estimation accuracy of 84.32% at run time.[1]

1 Introduction

Humans have an awareness of the scope and limit of their own knowledge (Fleming and Dolan, 2012; Koriat, 1997; Hart, 1965), as illustrated in Fig. 1. This cognitive self-awareness ability in humans introduces hesitation in us before we respond to queries or make decisions in scenarios where we know we don't know (Yeung and Summerfield, 2012; Nelson, 1990; Bland and Schaefer, 2012). However, LLM-based AI assistants lack this cognitive uncertainty estimation. Consequently, they tend to be overconfident and may produce plausible-sounding but unfaithful or nonsensical contents called *hallucination* or *confabulation* (Ji et al., 2022; Xiao and Wang, 2021; Bang et al., 2023; Xiong et al., 2023). This problem limits their applications in numerous real-world scenarios and undermines user trustworthiness.

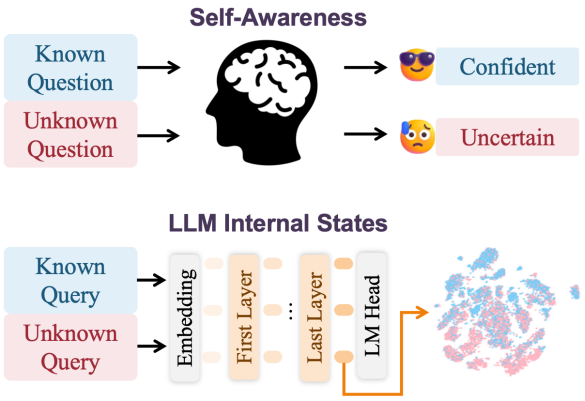


Figure 1: Humans have self-awareness and recognize uncertainties when confronted with unknown questions. LLM internal states reveal uncertainty even before responding. Pink dots are the internal LLM states associated with hallucinated responses, whereas Blue dots are those of faithful responses. The queries leading to those LLM responses are colored accordingly.

Previous research (Bricken et al., 2023; Templeton et al., 2024; Bills et al., 2023; Wu et al., 2024) have explored the internal states of language models that capture contextual and semantic information learned from training data (Liu et al., 2023; Chen et al., 2024; Gurnee and Tegmark, 2023). Nevertheless, internal states of language models sometimes exhibit limited generalization on unseen data and their representation effectiveness can be undermined by flawed training data or modeling issues (Wang et al., 2022a; Belinkov and Glass, 2019; Meng et al., 2021; Xie et al., 2022; Carlini et al., 2021; Yin et al., 2023a). Notably, recent works have shown that the LLM's internal states can potentially detect hallucinations in texts (Azaria and Mitchell, 2023; Chen et al., 2024; Su et al., 2024). However, these works examine texts not exclusively produced by the same LLMs whose internal states are analyzed, highlighting the necessity for further investigation into the LLM *self-awareness*

[1]The source code can be obtained from https://github.com/ziweiji/Internal_States_Reveal_Hallucination

and how their internal states correlate with their uncertainty and *own* hallucination occurrence.

Our work takes a step further by investigating **whether LLM internal states have some indication of hallucination risk given queries and whether it can be reliably estimated even before the actual response generation** (Fig. 1). We conduct a comprehensive analysis of LLMs internal mechanisms in terms of training data sources and across 15 diverse NLG tasks that extend beyond the QA task (Snyder et al., 2023; Slobodkin et al., 2023) and span over 700 datasets. We explore particular neurons, different activation layers, and tokens that play a crucial role in the LLM perception of uncertainty and hallucination risk. Employing a probing estimator (Belinkov, 2022) on the internal states associated with the queries, we validate their self-awareness and ability to indicate uncertainty in two aspects: (1) Whether they have seen the query in training data, achieving an accuracy of 80.28%. (2) Whether they are likely to hallucinate regarding the query, achieving an average estimation accuracy of 84.32% across 15 NLG tasks. We propose that understanding these representations could offer a proactive approach to estimating uncertainty, potentially serving as an early indicator for the necessity of retrieval augmentation (Wang et al., 2023) or as an early warning system.

2 Hallucination and Training Data

The sources of hallucination in LLMs can be traced back to *data* and *modeling* (Ji et al., 2022). Factors tied to data encompass unseen knowledge, task-specific innate divergence, noisy training data, etc. From the modeling perspective, hallucinations can be traced to the model architecture, alignment tax, teacher-forced maximum likelihood estimation (MLE) training, etc.

A common situation where hallucinations from *data* occur is when LLMs attempt to provide information on **unseen** queries that are not included in their training set, rather than refusing to reply. Previous works (Kadavath et al., 2022; Rajpurkar et al., 2018; Onoe et al., 2022; Yin et al., 2023b) explore to identify the unseen data based on various indicators such as text similarity, perplexity. We investigate the capability of LLMs to recognize **whether they have seen the query in training data** via novel analysis of their internal states. To facilitate analysis, we craft two sets of queries by collecting news from periods *before* and *after* the release of the LLM we analyze to represent unseen and seen data, respectively.

However, in the real-world scenario, it's impractical to definitively categorize data as entirely seen or unseen due to the inability to access the vast training data of LLM. Thus, we expand the preliminary insights and further investigate LLMs' self-awareness of recognizing **whether models are likely to hallucinate regarding the query.** It's important to note that the hallucinations are *source-agnostic*, meaning they can result from both unseen and seen data. The latter can still trigger hallucinations due to deficiencies in *modeling*. To facilitate analysis, we construct data by using LLM to directly generate responses to queries across diverse NLG tasks and then label the hallucination level in the responses.

3 Methodology

This section begins with an introduction to the problem formulation of uncertainty estimation faced with queries in § 3.1. We construct datasets in § 3.2 focusing on two dimensions: (1) the distinction between queries seen and unseen in the training data; (2) the likelihood of hallucination risk faced with the queries. To validate the efficacy of internal state representation in hallucination estimation, we visualize the neurons for perception extracted from a specified LLM layer (§ 3.3) and then leverage the *probing classifier* technique (Belinkov, 2022) on top of internal states associated with the last token of queries (§ 3.4).

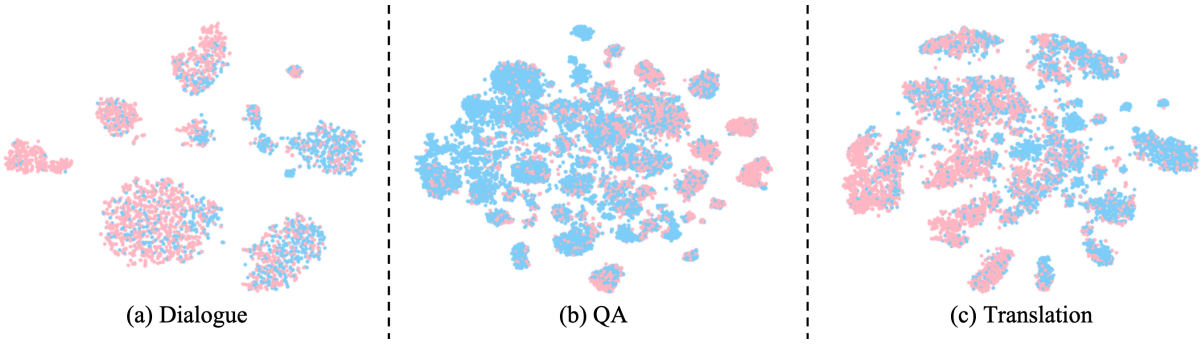
3.1 Problem Formulation

Suppose we have an LLM f parameterized by θ. It is able to gain internal states I and generate response r given user query q represented as $I_{\theta,q}, r_{\theta,q} = f_\theta(q)$. We aim to investigate the *self-awareness* of LLM, specifically how their internal states I relate to their level of hallucination risk h when faced with a query q.

We employ a dataset $\mathcal{D}_\theta = \{\langle I^{\text{train}}_{\theta,q,i}, h^{\text{train}}_i \rangle\}_{i=1}^{N}$ consisting of N query-label pairs. These pairs serve to represent the behavior of f_θ Here, h^{train}_i denotes the level of hallucination risk, which is labeled based on (1) the query's presence in the training data or (2) the degree of hallucination in the response r to q. Thus, our objective is mathematically expressed as:

$$h = \mathbb{E}(I_{\theta,q}; \mathcal{D}_\theta) \quad (1)$$

Here, $\mathbb{E}$ signifies an estimator function. The combi-



(a) Dialogue (b) QA (c) Translation

Figure 2: Visualization of the Neurons for Hallucination Perception in various NLG tasks. Pink dots represent Unknown Queries triggering hallucinations and Blue dots represent Known Queries.

nation of *model-specific* attribute via f_θ, and *query-only* attribute via q, allows it to accurately capture the characteristics of individual LLMs and fosters a more efficient prediction mechanism that mirrors human cognitive processes.

3.2 Data Construction

As introduced in § 1 and 2, we investigate LLM internal states' self-awareness and ability to indicate uncertainty in two aspects: (1) whether they have seen the query in training data; and (2) whether they are likely to hallucinate when faced with the query.

(1) Seen/Unseen Query in Training Data The unseen queries will trigger hallucinations due to the lack of information within the model's training data when the model doesn't refuse to respond. In other words, hallucinations triggered by unseen queries are *data*-related. To investigate the distinguishability between seen and unseen queries, we construct a compact dataset consisting of two distinct sets of queries. For the seen group, we utilize historical BBC news in 2020 highly likely exposed during LLM's training. For the unseen group, we utilize recent BBC news in 2024 *after* the release of the LLM we analyze. To ensure comparability, we ensure these two sets share similar length distributions and semantic information via sentence embeddings [2]. The two groups are both from LatestEval (Li et al., 2024), a benchmark designed to tackle data contamination in evaluation through dynamic and time-sensitive construction. The query is `Tell more details about the news: {news_title}`.

[2] https://huggingface.co/sentence-transformers/all-mpnet-base-v2

(2) Hallucination Risk faced with the Query We first construct data using LLM to directly generate responses to queries in diverse NLG tasks. Subsequently, for labeling the responses and corresponding queries, a comprehensive integration of NLG metrics assesses the levels of hallucination.

We select 15 NLG task categories including QA, Summarization, Translation, etc consisting of over 700 datasets from **Super-Natural Instructions** benchmark (Wang et al., 2022b) [3]. This generation only uses the parametric knowledge of LLM which is a proxy of performance in real-world applications. This generation process can also be in other settings, such as retrieval-augmented generation (RAG), to explore whether the internal states can estimate hallucination risk or other aspects' performance in these settings.

To evaluate the generated responses, we implement a multi-faceted evaluation approach. We employ classical **Rouge-L** (Lin, 2004), which compares the generated response with gold-standard reference. To measure hallucination level, we also utilize **Natural Language Inference (NLI)** [4] and **Questeval** (Scialom et al., 2021). **NLI** is a common metric for hallucination evaluation (Ji et al., 2023a,b) which assesses the logical consistency/entailment of generated text with the provided context or the reference. **QuestEval** is a QA-based metric for evaluating the faithfulness of the output in generation tasks. This work adopts its reference-dependent mode depending both on the input source and golden reference.

To make up for deficiencies of single automatic metrics, we integrate these three metrics compre-

[3] Please find the full list of NLG task categories in Fig. 3 or 4 and the full list of tasks in Tab. A2 in § A.

[4] https://huggingface.co/MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7

hensively. If NLI predicts entailment and both Rouge-L and Questeval exceed their respective median values, we assign a label of **1**. Conversely, if NLI predicts contradiction or neutrality, and both Rouge-L and Questeval fall below their median values, we assign a label of **0**. This labeling strategy not only provides a binary quality assessment but also reflects a multi-dimensional evaluation of the text, capturing the hallucination level of the generated responses.

3.3 Preliminary Analysis: Neurons for Hallucination Perception from Internal States

Internal states play a crucial role in language models, encapsulating rich contextual and semantic information learned from predicting tokens. They are adept at recognizing complex patterns and relationships pertinent to various NLP tasks, which positions them as potentially powerful tools for estimating the risk of hallucinations (Azaria and Mitchell, 2023; Liu et al., 2023; Chen et al., 2024). Furthermore, previous works (Azaria and Mitchell, 2023; Ahdritz et al., 2024; Liu et al., 2024) have demonstrated that the activations of the last token from the last layer in LLMs contain one of the most useful features. Therefore, we take these representations for preliminary analysis on the self-assess sense of internal states and the role of specific neurons in the uncertainty and hallucination estimation. Specifically, we employ a feature selection method based on Mutual Information (Kraskov et al., 2004) to measure the relevance of different features/dimensions for distinguishing between the categories in a dataset.

In the context of NLG tasks including dialogue, QA, and translation, we select the eight most significant neurons/dimensions from the last activation layer and visualize them in Fig. 2. We observe that these neurons exhibit sensitivity to uncertainty, allowing them to distinguish between different hallucination levels given known and unknown queries. In other words, there exist individual neurons within LLM that can fairly perceive uncertainty and predict future hallucinations. This approach not only enhances our understanding of the neural correlates of hallucinations but also paves the way for developing targeted interventions that mitigate the effects of hallucinations.

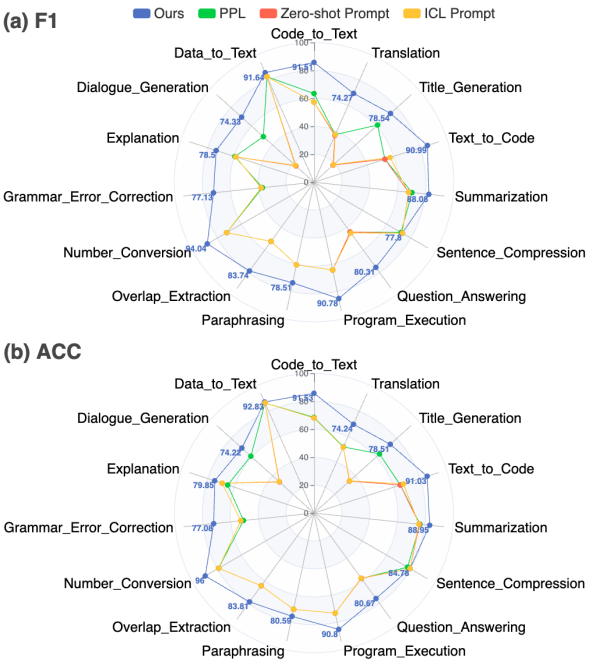


Figure 3: Automatic evaluation results for our method and baselines including Perplexity (PPL), Zero-shot Prompt, and In Context Learning (ICL) Prompt.

3.4 Internal State-based Estimator

Based on the above preliminary analysis and previous works, we use the activations corresponding to the last token of queries from a specified layer in LLMs, denoted as x_q, as the input for our estimation model. The accessibility and ease of obtaining these states further underscore their practicality for such applications.

For the architecture of our estimator, we employ a variant of the multilayer perceptron (MLP) adapted from the Llama (Touvron et al., 2023). The estimator is mathematically formulated as:

$$H = \text{down}(\text{up}(x_q) \times \text{SiLU}(\text{gate}(x_q))) \quad (2)$$

where SiLU is the activation function. down, up, and gate are linear layers for down-projection, up-projection, and gate mechanisms, respectively. The combination of internal states and the Llama MLP structure handles the complexity of hallucination risk estimation in NLG tasks.

4 Experiments

4.1 LLM

In this work, we primarily use Llama2-7B (Touvron et al., 2023) as our generative model and delve into its internal states to access hallucination risk estimation. In addition, we explore the impact of